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香港科技大學
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AND TECHNOLOGY

Deep Learning-enabled Prediction of Intrinsically Disordered Protein Regions

CHEM4420

Statistical Machine Learning Methods for Chemical Data Analysis

7 May 2026

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1. Introduction of Intrinsically Disordered Proteins (IDPs)

Unlike globular proteins, Intrinsically Disordered Proteins (IDPs) consist of proteins or regions that lack a stable three-dimensional structure. (Trivedi & Nagarajaram, 2022). They are involved in important processes like cell signalling and molecular assembly (Wright & Dyson, 2014) while remaining unfolded. Their ever-changing forms present difficulty in identifying disordered regions with traditional biological methods. While recent advancements such as AlphaFold 2 and AlphaFold 3 can predict a protein's three-dimensional structure, these softwares still cannot predict disorderiness in proteins. This report thus aims to explore the usage of machine learning methods to predict intrinsically disordered regions within protein sequences.

1.1. Background of IDPs

Beyond their crucial roles in cell signaling and regulation, they are also linked to neurological diseases such as Alzheimer's due to their ability to undergo liquid-liquid phase separation (LLPS) (Utami et al., 2025). This is possible due to low sequence complexity, formed from repeated amino acids that form weak cross-links, driving the separation of molecules and the eventual formation of solid aggregates (Dinic et al., 2021) that are toxic to the cell. This highlights the importance of being able to predict IDPs beyond the current scope of traditional methods.

1.2. Experimental Characterization of IDPs

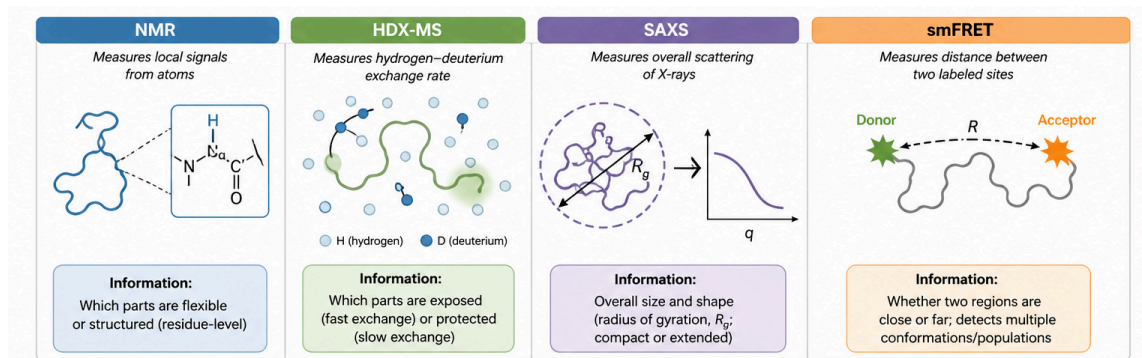


Figure 1. Overview of experimental methods used to characterize IDPs

Conventional methods like X-ray crystallography are not suitable for IDPs because they lack a stable three-dimensional fixed structure. Instead, solution-based methods such as nuclear magnetic resonance (NMR) spectroscopy, hydrogen-deuterium exchange mass spectrometry (HDX-MS), small-angle X-ray scattering (SAXS), and single-molecule Förster resonance energy transfer (smFRET) are used to study their dynamic behaviour. As illustrated in Figure 1, each method describes a different aspect of protein structure, including local flexibility, solvent exposure, overall size, and long-range distances.

1.3. Limitations of Experimental Methods

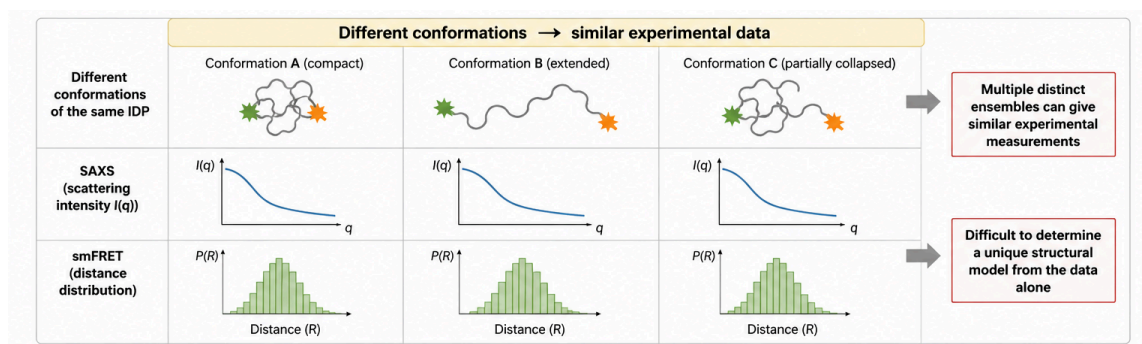


Figure 2. Underdetermination in IDPs

Although these methods are complementary, each provides only partial information about the conformational ensemble of IDPs. Figure 2 illustrates a key conceptual challenge, known as the

problem of underdetermination, where multiple conformational ensembles can yield similar experimental results, making it difficult to determine a unique structural model.

Many methods also rely on ensemble-averaged measurements. NMR and SAXS methods capture signals averaged over large populations of molecules, which might mask transient or low-population conformations that may be functionally important (Dyson & Wright, 2021).

However, each method has practical limitations. NMR is time-consuming and requires high sample concentrations (Dyson & Wright, 2021), whereas HDX-MS provides limited spatial resolution (Balasubramaniam & Komives, 2013), SAXS lacks residue-level structural detail (Cordeiro et al., 2017), and smFRET is restricted to measurements between selected labelled sites (Gomes et al., 2017). Furthermore, experimental datasets are often incomplete, especially for synthetic proteins or poorly annotated proteins.

These limitations highlight the challenges of fully characterizing IDPs using experimental methods alone. As a result, machine learning models have been developed to complement experimental data. In this project, we propose a machine learning model to address these challenges, which will be discussed in the following section.

2. Computational and Machine Learning Approaches

2.1. Prior Methods

Machine learning methods to predict intrinsically-disordered regions of proteins have been developed which utilize a variety of data sources and embedding architectures, although such methods are usually met with specific flaws that limit its scope. Firstly, given how intrinsically-disordered regions are challenging to characterize through experimental methods

with high-efficiency, simulation based methods have been provided as an alternative to evaluate the “disorderness” of a protein. In fact, deep learning models trained on simulated protein features have already been developed which reliably maps several features of a protein sequence into a regressor predicting radius of gyration (Rg), heat capacity, decorrelation time, and other features (Mollaei et al., 2024). However, such models do not provide discrete information about the specific regions that a protein could be disordered, hindering understanding of its biochemical activity. Other methods to predict intrinsically-disordered regions of proteins using deep neural networks involving several inputs such as co-evolutionary information (ex. Protein-specific substitution matrix (PSSM) from BLAST search) (Hanson et al., 2019) or processed information from NMR/crystal structure data (ex. DSSP notation of structure) (Hou et al., 2024). However, such approaches are limited to the richness of high-quality data sources required, restricting dataset size and increasing overfitting. Furthermore, multi-modal data sources often require normalization steps before usage, limiting applicability for novel predictions while also facilitating issues with feature loss. With an influx of datasets covering intrinsically-disordered protein regions available, coinciding with the rise of attention-based architectures and more recently biological sequence-based large language models, a more direct method is needed to intuitively predict protein regions that are intrinsically disordered.

2.2. Data Collection

The dataset used for this report was primarily taken from three separate datasets:

- (1) Disprot (3199 sequences obtained, Version 9.9 - last updated 2025.12) - A general databank of proteins with intrinsically-disordered regions from several organisms with different functions, annotations derived from a consensus of experimental results.

Link: <https://disprot.org/>

- (2) LLPSDB (332 sequences, last updated 2019.7) - A databank of proteins undergoing liquid-liquid phase separation, each sequence contains several studies that describe propensity to change phase, mix between naturally-derived and designed proteins.

Link: <http://bio-comp.org.cn/llpsdb/home.html>

- (3) FuzDB (372 sequences) Database of protein-biomolecule interactions where the interaction site is conformationally heterogeneous or “fuzzy”. While it can overlap with PDB, Disprot, or LLPSDB entries, FuzDB focuses on biomolecular interactions of proteins that involve intrinsically-disordered regions.

Link: <https://fuzdb.org/>

Each dataset was filtered for quality issues (ex. Length mismatch between IDR annotations and amino acid sequence) and cross-checked for duplicate entries (duplicate sequence-disorder annotation combinations were discarded)

2.3. Machine Learning Model - Direct Sequence-to-Disorder Region Approach

Our primary deep learning approach utilizes a sequence-to-disorder region approach where a tokenized amino acid sequence is directly mapped . Such approach was considered feasible due to the sequence-based nature of (either through patterns - intrinsically-disordered regions

2.4. Technical Enhancements & Post-processing

To enhance model robustness and biological relevance, the framework integrates specialized loss functions, performance metrics, and post-processing refinements. The optimization strategy utilizes Focal Loss to address class imbalance by adjusting the loss contribution of easy examples via a focusing parameter, forcing the model to prioritize the 19.8% minority disordered class. Simultaneously, Total Variation (TV) Regularization is incorporated to enforce local

continuity and penalize high-frequency noise, ensuring predicted regions remain biologically contiguous. Model performance is monitored via the Macro F1-score, which provides an unweighted average of class-specific F1-scores to prevent evaluation bias toward the majority ordered class.

Following prediction, sequence smoothing is performed using independent filtering techniques to ensure structural realism. Depending on the noise profile, individual applications of Average, Median, or Savitzky-Golay filtering—the latter performing local polynomial regression—are employed to suppress isolated spikes and local variance. These filters are supplemented by a minimum length constraint (e.g., < 3 residues) to eliminate spurious noise and align with the physical nature of disordered regions. Finally, rather than adhering to a default 0.5 cutoff, the decision threshold is optimized through 5-fold cross-validation to maximize the Macro F1-score and ensure effective generalization to unseen data.

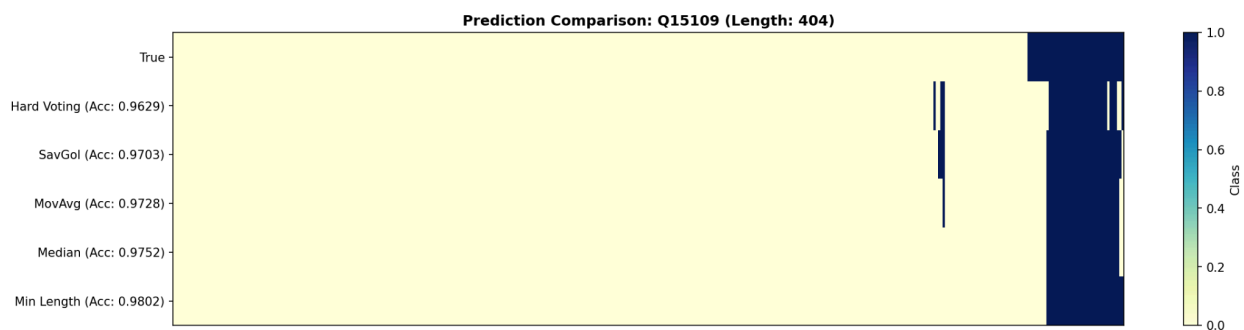


Figure 3. Visualization of Prediction Post-processing and Smoothing for a query protein with Uniprot ID Q15109. (add: should we actually say what type of protein this is, describe why the

C-terminal is disordered)

2.5. ESM2-based Machine Learning Approaches for IDP Prediction

Recent developments have leveraged protein language models like ESM-2 to advance computational prediction of intrinsically disordered protein regions. DisPredict3.0, for instance, leverages the ESM-2 protein language model and significantly outperforms existing methods in multiple metrics (Kabir et al., 2024). Similarly, DeepDRP utilizes advanced deep learning methodologies, integrating transformer-enhanced embeddings, particularly ESM-2, to achieve superior prediction of intrinsically disordered proteins (Yang et al., 2023). The IDP-ELM method demonstrates improved accuracy, efficiency, and convenience in identifying intrinsically disordered regions and their biological functions by leveraging high-dimensional representations from state-of-the-art protein language models and ensemble learning (Xu & Onoda, 2023).

2.6. Tree-Based Machine Learning Algorithms

XGBoost is a scalable tree-boosting system that incorporates novel techniques, including a sparsity-aware algorithm and a weighted quantile sketch, as well as optimizations for memory and computation (Chen & Guestrin, 2016). These advances enable XGBoost to handle massive datasets with improved resource efficiency while delivering state-of-the-art performance. Similarly, LightGBM introduces innovative techniques such as Gradient-based One-sided Sampling (GOSS) and Exclusive Feature Bundling (EFB) to significantly enhance training speed and reduce memory usage (Ke et al., 2017). These optimizations make LightGBM particularly effective for large-scale machine learning tasks. Furthermore, CatBoost addresses the "prediction shift" problem caused by target leakage in gradient boosting algorithms through the novel implementation of ordered boosting and an advanced algorithm for handling categorical features (Prokhorenkova et al., 2018). These innovations result in superior performance and empirical

quality compared to other boosting implementations across diverse biological and computational datasets.

2.7. Hybrid Approach: Combining PLMs with Tree-Based Models

The proposed hybrid approach leverages the synergy between ESM-2's deep evolutionary "grammar" and the superior ability of tree-based models to handle non-linear tabular data. As illustrated in the workflow, raw sequences from the training and test sets are first processed through the ESM-2 model to extract high-dimensional embeddings. This process captures complex latent biological features automatically, eliminating the need for manual calculation of physical properties. These embeddings then serve as the input features for a suite of tree-based models, specifically XGBoost, LightGBM, and CatBoost, which are trained to define sharper decision boundaries for Order vs. Disorder classification. To further enhance robustness, the individual predictions from these models are aggregated through an ensemble layer using Hard or Soft Voting to produce the final classification. This pluggable architecture not only allows for easy swapping of ESM-2 with other LLMs like ProtT5 or Ankh to optimize for different structural nuances but also utilizes the inherent parameters of Gradient Boosting Trees to effectively manage class imbalance.

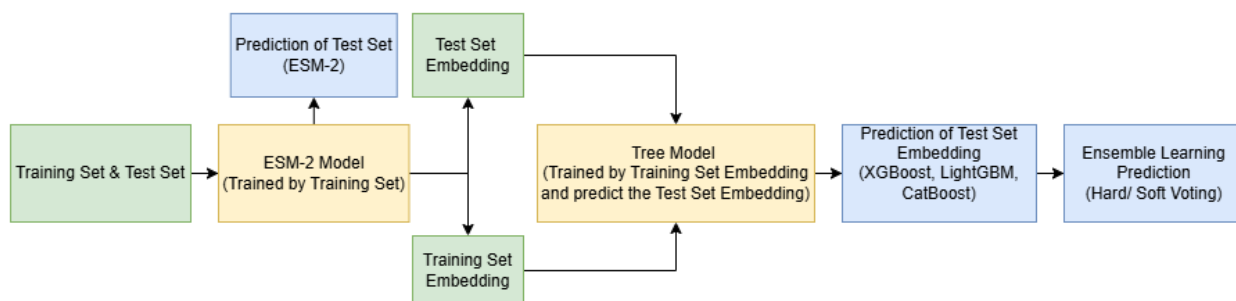


Figure 4. Workflow for the Hybrid Approach

2.8. Performance Comparison of ESM-2 and Tree-Based Hybrid Models

The comparative analysis reveals that integrating the ESM-2 protein language model with tree-based gradient boosting algorithms significantly improves predictive performance over a standalone ESM-2 model. While the baseline ESM-2 achieves a Test Macro F1-score of 0.67, all three tree-based implementations—XGBoost, LightGBM, and CatBoost—consistently outperform it, each reaching a score of 0.70 on the test set.

Among the individual models, CatBoost demonstrates the highest training robustness with a 5-fold CV Macro F1-score of 0.759, followed by LightGBM (0.737) and XGBoost (0.731). Despite the variations in cross-validation, all tree models converge to an identical Test Macro F1 of 0.70, suggesting a performance ceiling for individual classifiers on this specific dataset. Furthermore, employing Ensemble methods such as Soft and Hard Voting maintains this high performance (0.70), though they do not yield additional gains over the individual tree models. The optimized thresholds (ranging from 0.66 to 0.68) are notably higher than the default 0.5, indicating that shifting the decision boundary was essential for maximizing the Macro F1-score and addressing the underlying class imbalance.

2.9. Key Advantages and Potential

The hybrid framework of protein language models and gradient boosting trees offers several strategic advantages for biological sequence analysis. First, the architecture excels at handling class imbalance, a ubiquitous challenge in IDP datasets where disordered residues are the minority; tree models provide native parameter support to effectively manage this skew. Second, the pluggable nature of LLMs ensures future-proofing; the modular design allows for the seamless swapping of ESM-2 with newer models like ProtT5, ProtBERT, or Ankh to optimize for specific structural nuances or larger sequence contexts. Finally, this approach offers classifier

flexibility, as the high-dimensional embeddings act as a universal feature set that can be paired with any downstream head—ranging from Random Forest and SVM for smaller datasets to MLPs for complex, large-scale deep learning tasks—depending on the specific research requirements.

Challenges and Future Outlook

3.1. Challenges

a. Data Quality - The “Ground Truth” Ambiguity

The definition of "disordered" lacks standardization across experiments, resulting in a spectrum of loose terms (Dunker, A. K. et al., 2013). This has created a "consistency gap" (Shi, Peng; et al., 2019), where identical proteins yield conflicting results due to differing analytical tools used: X-Ray Crystallography identifies disorder by missing electron density in crystal structures (Zanotti, G. 2023), whereas NMR often produces overlapping signals between "highly flexible" and "completely disordered" regions (Felli, I. C., & Pierattelli, R. (Eds.). 2015). Such methodological differences make it difficult to integrate data from major databases like DisProt. To resolve this ontological issue, DisProt recently adopted the MIADE (Minimum Information About Disorder Experiments) guidelines (Nugnes, Maria Victoria et al., 2026), a significant step toward consistent data documentation.

b. Nature of IDPs - Environmental and Contextual Sensitivity

Predictive models often overlook the “Shapeshifter” nature of IDPs. Rather than a fixed state, IDPs exist as conformational ensembles highly sensitive to their surroundings (Chakravarty, D., Schafer, J. W., & Porter, L. L. 2023). Current models rely on protein sequences alone, ignoring critical cellular factors:

pH: Changes in pH can neutralize these stabilizing charges, allowing hydrophobic forces to cause proteins to "collapse" (Lindsay, R. J., Mansbach, R. A. et al., 2021). Proteins labeled "disordered" at standard experimental pH of 7.4 may be ordered under actual cell conditions.

Molecular Crowding: IDPs analyzed in isolation do not reflect actual cellular environment. Intracellular molecules create “exclusion volumes” that can physically compress IDPs, sometimes inducing folding (Cino, E. A. et al., 2012). Advancements like DispHred (Jaime Santos et al., 2020) evaluates disorder as a function of pH, paving the way for more accurate simulations of IDPs within real-world biological contexts.

3.2. Future Outlook (Applications of Project)

a. Cancer Research - Targeting the “Undruggable”

Many critical cancer-related proteins, such as c-Myc, p53, and EWS-FLI1, are primarily disordered. Their lack of a stable "lock" structure makes it difficult for pharmaceutical companies to design specific "keys" required for traditional drug targeting (Barber-Rotenberg, et al., 2010). However, our model can be optimized to identify Molecular Recognition Features (MoRFs)—short disordered segments that stabilize upon binding to a partner. By pinpointing these MoRFs, researchers can design Inhibitors or Peptidomimetics to block these interaction motifs, effectively "switching off" the drivers of cancer (Malhis, N., & Gsponer, J. 2015).

b. Neurodegenerative Diseases - Preventing the “clump”

Protein aggregation is a hallmark of most neurodegenerative diseases, occurring when healthy, flexible proteins misfold into toxic, solid-like clumps (Pedersen, J. T et al., 2013). Proteins like Tau (Alzheimer’s) and Alpha-synuclein (Parkinson’s) naturally exist in disordered states but can aggregate into harmful structures. Our prediction model could allow researchers to pinpoint the

exact amino acids responsible for this clumping. Early prediction stabilizes specific residues, preventing toxic protein aggregation from occurring (KeAi Communications Co Ltd, 2024).

3. Conclusion

Experimental methods such as NMR, HDX-MS, SAXS, and smFRET give complementary insights into the structure and dynamics of IDPs. However, each method only captures a part of the conformational ensemble and is restricted by factors such as ensemble averaging, resolution, and experimental restrictions.

*martin and tim's part

The study of IDPs is challenged by inconsistent definitions across analytical methods and the "shapeshifter" nature of proteins that react to cellular pH and molecular crowding. However, optimising our predictive model can overcome these hurdles by identifying MoRFs to target "undruggable" cancer drivers and pinpointing specific amino acids to prevent the toxic protein clumping seen in Alzheimer's and Parkinson's.

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5. Project Contribution

Team Members	Contribution
Wang Martin Shen (20895152)	1. Developed the main dataset used in this project (data obtaining, processing, and filtering) 2. Designed and implemented “Direct sequence-to-disorder region” model 3. Wrote visualization methods to evaluate results + make predictions 4. Completed Sections 2.1-2.2
Tam Wing Ching (21007532)	1. Designed and implemented ‘ESM-2 and Tree-Based Hybrid Models.’

	2. Optimization (Focal Loss with TV regularization + Smoothing output, Rescheduler, Hyperparameter Tuning, Threshold Searching) 3. Completed Sections 2.3-2.6
Teo Xin Yi (21335501)	1. Contributed to brainstorming and development of the project topic 2. Identified and sourced relevant databases 3. Conducted data cleaning 4. Completed Sections 1.2. Experimental Characterisation of IDPs and 1.3. Limitations of Experimental Methods
Wang Qian Wen (21335537)	1. Contributed to brainstorming and development of the project topic 2. Identified and sourced relevant databases 3. Conducted data cleaning 4. Completed Sections 1. Introduction of Intrinsically Disordered Proteins (IDPs) and 1.1. Background of IDPs
Teng Jia Han (21335525)	1. Conducted data cleaning 2. Completed Section 3. Challenges and Future Outlook

Turnitin similarity report